

The background of the slide is a light gray gradient. It is decorated with several realistic water droplets of various sizes, some clustered in the top left and bottom right corners. In the center background, there is a faint, circular logo or watermark that is not clearly legible.

Operations on Fuzzy Sets

Basic fuzzy set operations: Union

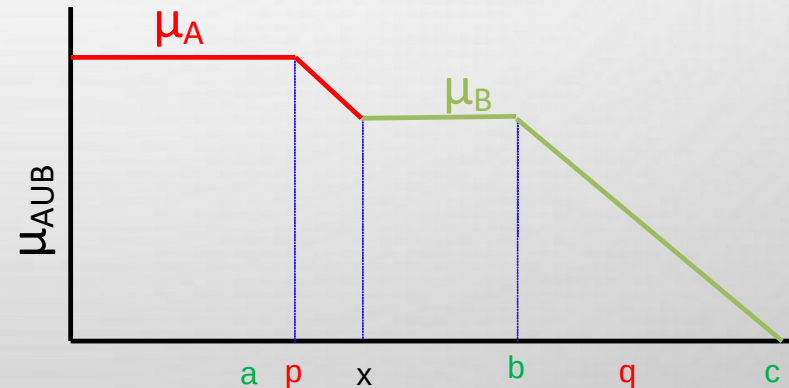
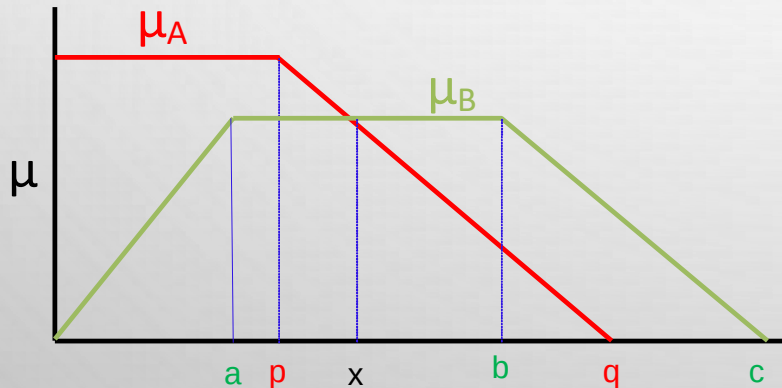
Union ($A \cup B$): $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$

Example:

$A = \{(x_1, 0.5), (x_2, 0.1), (x_3, 0.4)\}$ and

$B = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5)\};$

$C = A \cup B = \{(x_1, 0.5), (x_2, 0.3), (x_3, 0.5)\}$



Basic fuzzy set operations: Intersection

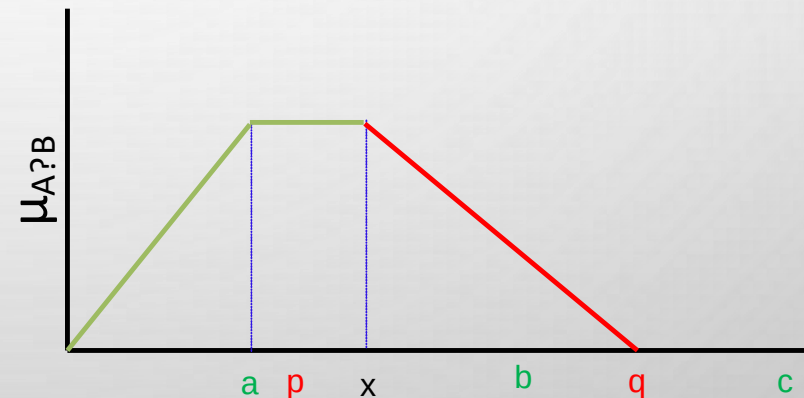
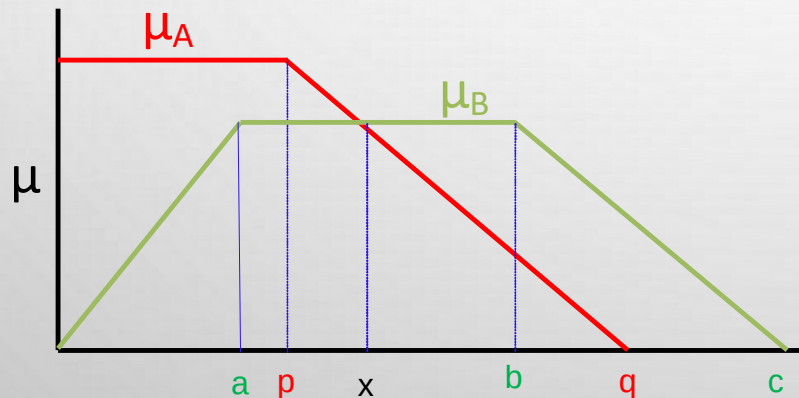
Intersection ($A \cap B$): $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$

Example:

$A = \{(x_1, 0.5), (x_2, 0.1), (x_3, 0.4)\}$ and

$B = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5)\};$

$C = A \cap B = \{(x_1, 0.2), (x_2, 0.1), (x_3, 0.4)\}$



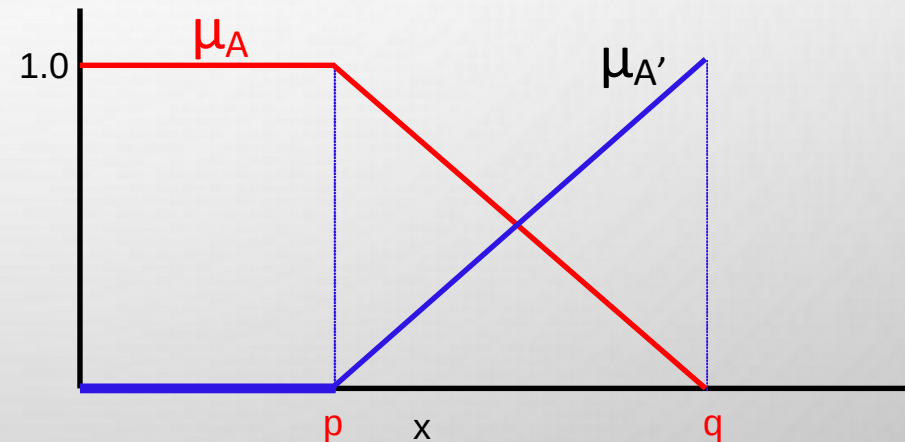
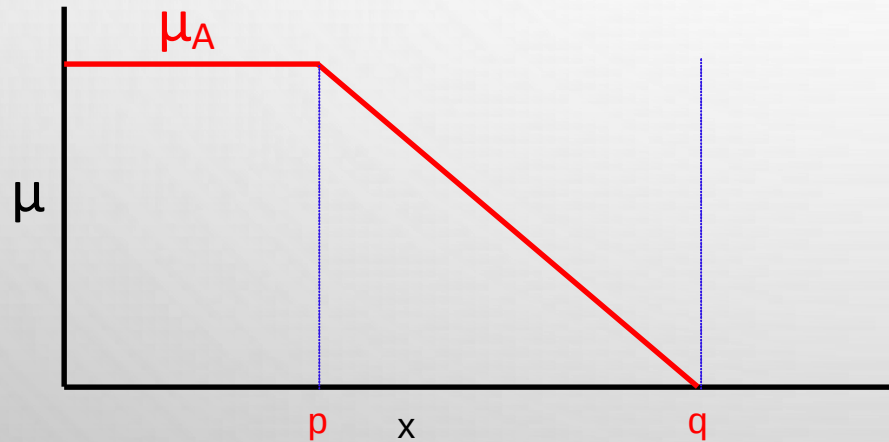
Basic fuzzy set operations: Complement

Complement (A^c): $\mu_{A^c}(x) = 1 - \mu_A(x)$

Example:

$$A = \{(x_1, 0.5), (x_2, 0.1), (x_3, 0.4)\}$$

$$C = A^c = \{(x_1, 0.5), (x_2, 0.9), (x_3, 0.6)\}$$



Basic fuzzy set operations: Products

Algebraic product or Vector product ($A \cdot B$):

$$\mu_{A \cdot B}(x) = \mu_A(x) \cdot \mu_B(x)$$

Scalar product ($\alpha \times A$):

$$\mu_{\alpha A}(x) = \alpha \times \mu_A(x)$$

Basic fuzzy set operations: Sum and Difference

Sum ($A + B$):

$$\mu_{A+B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x) \cdot \mu_B(x)$$

Difference ($A - B = A \cap B^C$):

$$\mu_{A-B}(x) = \mu_{A \cap B^C}(x)$$

Disjunctive sum:

$$A \oplus B = (A^C \cap B) \cup (A \cap B^C)$$

Bounded Sum:

$$|A(x) \oplus B(x)| = \mu_{|A(x) \oplus B(x)|} = \min\{1, \mu_A(x) + \mu_B(x)\}$$

Bounded Difference:

$$|A(x) \ominus B(x)| = \mu_{|A(x) \ominus B(x)|} = \max\{0, \mu_A(x) + \mu_B(x) - 1\}$$

Basic fuzzy set operations: Equality and Power

Equality ($A = B$):

$$\mu_A(x) = \mu_B(x)$$

Power of a fuzzy set A^α :

$$\mu_{A^\alpha}(x) = (\mu_A(x))^\alpha$$

- ✓ If $\alpha < 1$, then it is called **dilation**
- ✓ If $\alpha > 1$, then it is called **concentration**

Basic fuzzy set operations: Cartesian product

Cartesian Product ($A \times B$): $\mu_{A \times B}(x, y) = \min(\mu_A(x), \mu_B(y))$

Example:

$$A(x) = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5), (x_4, 0.6)\}$$

$$B(y) = \{(y_1, 0.8), (y_2, 0.6), (y_3, 0.3)\}$$

$$A \times B = \min(\mu_A(x), \mu_B(y)) = \begin{array}{c} \begin{matrix} & y_1 & y_2 & y_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} \left[\begin{array}{ccc} 0.2 & 0.2 & 0.2 \\ 0.3 & 0.3 & 0.3 \\ 0.5 & 0.5 & 0.3 \\ 0.6 & 0.6 & 0.3 \end{array} \right] \end{array}$$

Properties of fuzzy sets

Properties of fuzzy sets

Commutativity :

$$A \cap B = B \cap A$$

$$A \cup B = B \cup A$$

Associativity :

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributivity :

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Properties of fuzzy sets

Idempotence :

$$A \cup A = A$$

$$A \cap A = \emptyset;$$

$$A \cup \emptyset; = A$$

$$A \cap \emptyset; = \emptyset;$$

Transitivity :

If $A \subseteq B; B \subseteq C$ then $A \subseteq C$

Involution :

$$(A^c)^c = A$$

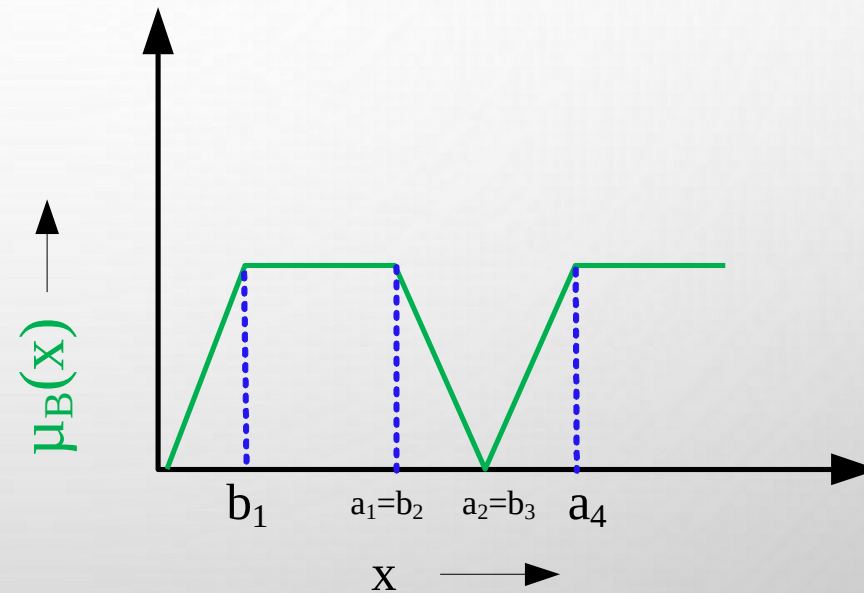
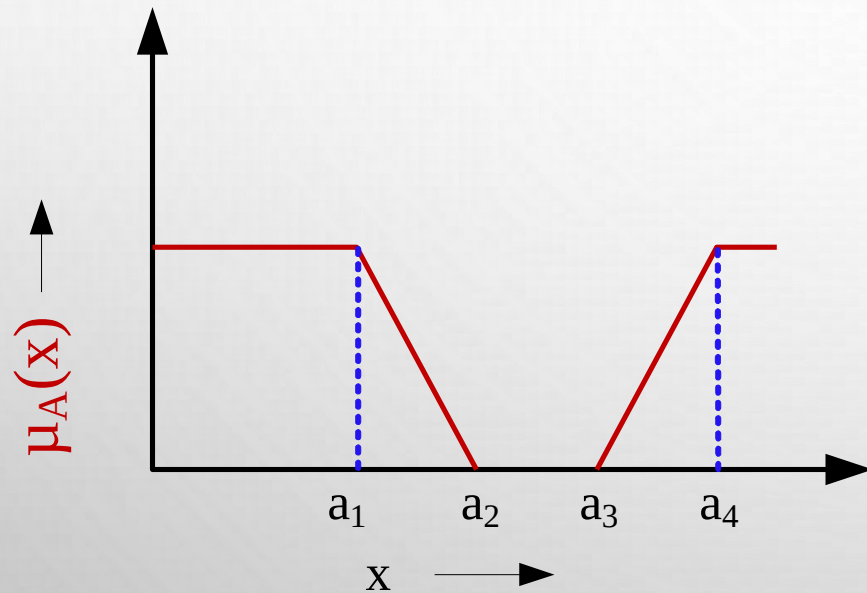
De Morgan's law :

$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

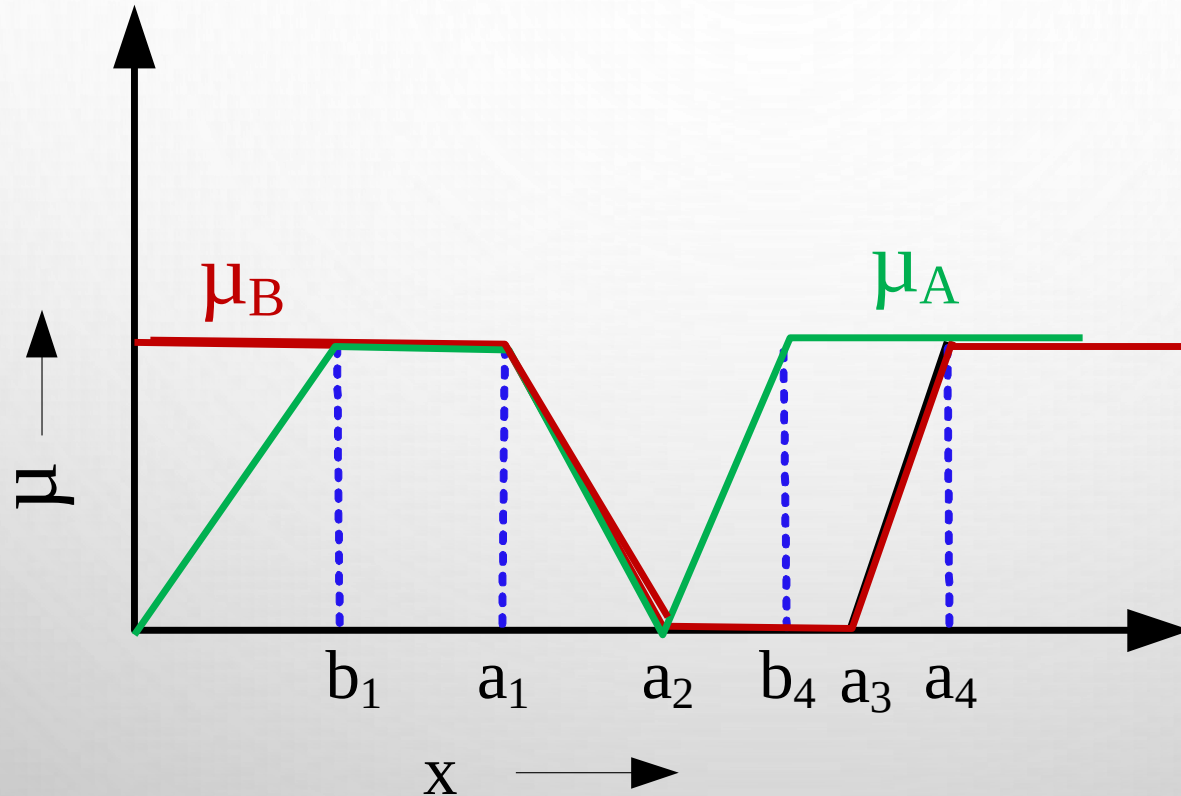
Example 1: Fuzzy Set Operations

Let A and B are two fuzzy sets defined over a universe of discourse X with membership functions $\mu_A(x)$ and $\mu_B(x)$, respectively. Two MFs $\mu_A(x)$ and $\mu_B(x)$ are shown graphically.



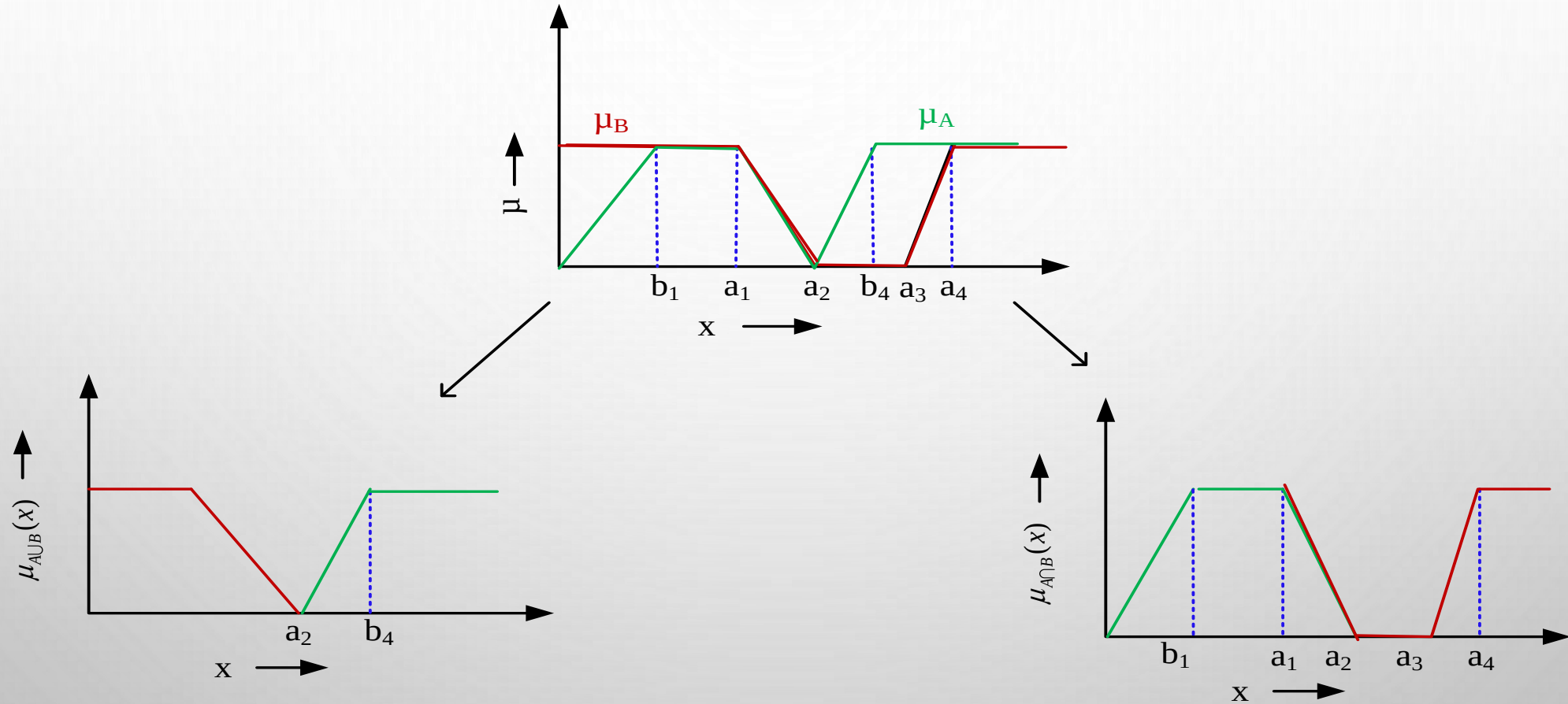
Example 1: Plotting two sets on the same graph

Let's plot the two membership functions on the same graph



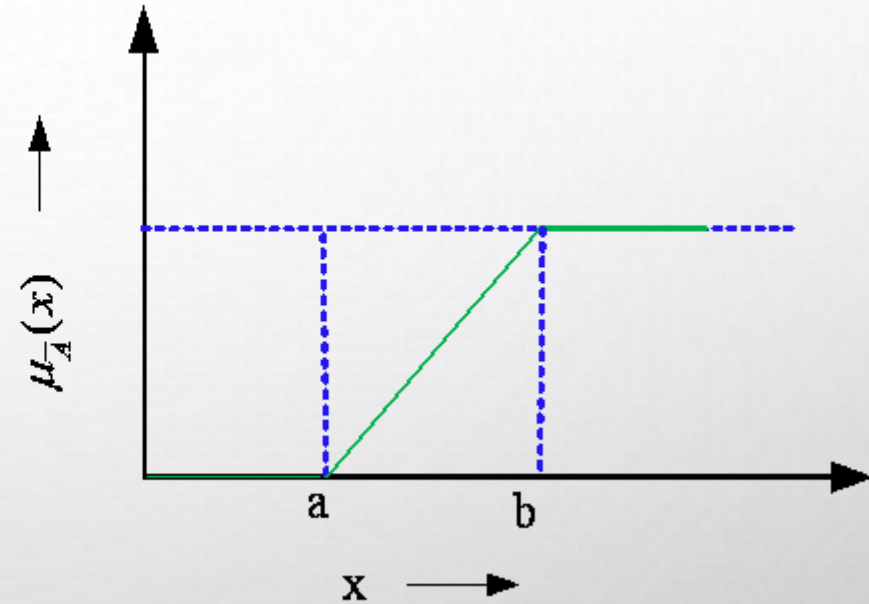
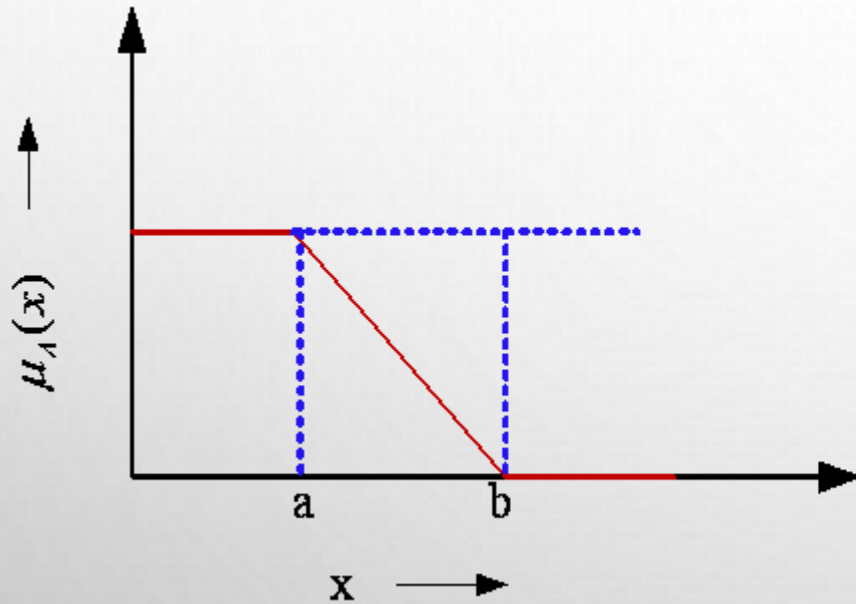
Example 1: Union and Intersection

The plots of union $A \cup B$ and intersection $A \cap B$ are shown in the following.



Example 1: Complementation

The plots of union $\mu_{\bar{A}}(x)$ of the fuzzy set A is shown in the following.



Fuzzy set operations: Practice

Consider the following two fuzzy sets A and B defined over a universe of discourse $[0,5]$ of real numbers with their membership functions

$$\mu_A(x) = \frac{x}{1+x} \text{ and } \mu_B(x) = 2^{-x}$$

Determine the membership functions of the following and draw them graphically.

I. $\bar{A}, \bar{B}$

II. $A \cup B$

III. $A \cap B$

IV. $(A \cup B)^c$

[Hint: Use De' Morgan law]

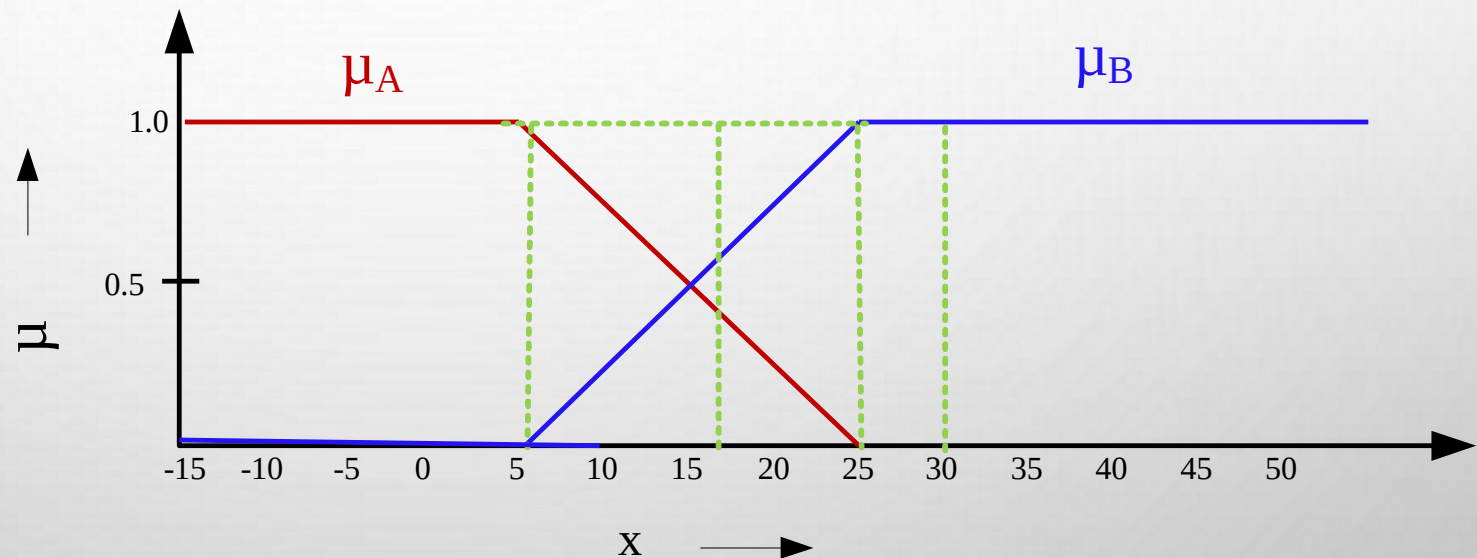
Example 2: A real-life example

Two fuzzy sets A and B with membership functions $\mu_A(x)$ and $\mu_B(x)$, respectively defined as below.

A = **Cold climate** with $\mu_A(x)$ as the MF.

B = **Hot climate** with $\mu_B(x)$ as the M.F.

Here, X being the universe of discourse representing entire range of temperatures.



Example 2: A real-life example

What are the fuzzy sets representing the following?

1. **Not cold climate**
2. **Not hot climate**
3. **Extreme climate**
4. **Pleasant climate**

Note: Note that "Not cold climate" $\neq$ "Hot climate" and vice-versa.

Example 2: A real-life example

Answer would be the following.

✓ **Not cold climate**

$\bar{A}$ with $1 - \mu_A(x)$ as the MF.

✓ **Not hot climate**

$\bar{B}$ with $1 - \mu_B(x)$ as the MF.

✓ **Extreme climate**

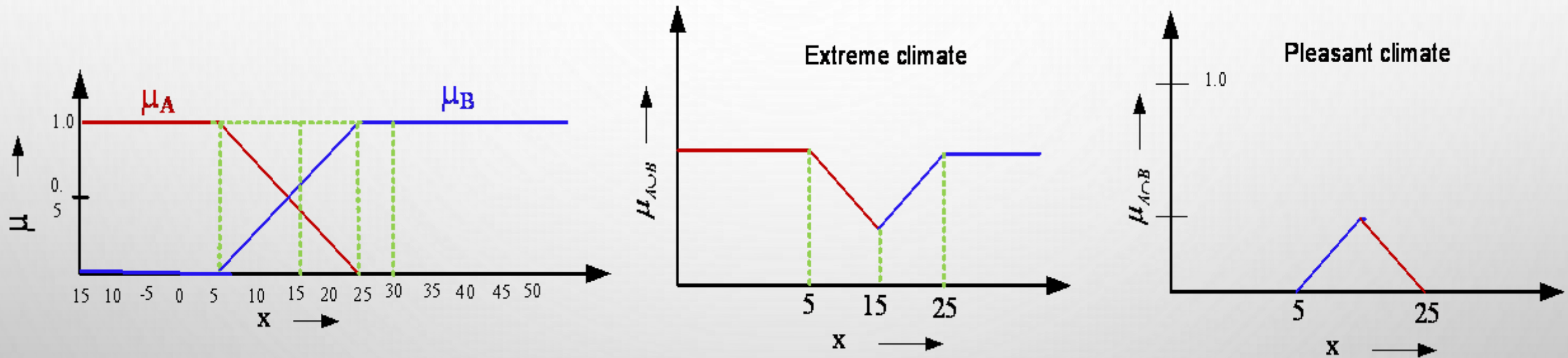
$A \cup B$ with $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$ as the MF.

✓ **Pleasant climate**

$A \cap B$ with $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$ as the MF.

Example 2: A real-life example

The plot of the MFs of $A \cup B$ and $A \cap B$ are shown in the following.



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Fuzzy Relations

Fuzzy Relations

- **Crisp relations**
- **Operations on crisp relations**
- **Examples on crisp relations**
- **Fuzzy relations**
- **Operations on fuzzy relations**
- **Examples on fuzzy relations**

Crisp relations

- **Order pairs:**

- Suppose, A and B are two (crisp) sets. Then Cartesian product denoted as $A \times B$ is a collection of order pairs, such that

$$A \times B = \{(a, b) | a \in A \text{ and } b \in B\}$$

Note :

- (1) $A \times B \neq B \times A$
- (2) $|A \times B| = |A| \times |B|$
- (3) $A \times B$ provides a mapping from $a \in A$ to $b \in B$.

A particular mapping so mentioned is called a **relation**.

Crisp relations

Example:

- Consider the two crisp sets A and B as given below.
 $A = \{1, 2, 3, 4\}$ $B = \{3, 5, 7\}$.

Then, $A \times B = \{(1, 3), (1, 5), (1, 7), (2, 3), (2, 5), (2, 7), (3, 3), (3, 5), (3, 7), (4, 3), (4, 5), (4, 7)\}$

Let us define a relation as $R = \{(a, b) | b = a + 1, (a, b) \in A \times B\}$

Then, $R = \{(2, 3), (4, 5)\}$ in this case.

Crisp relations

- We can represent the relation R in a matrix form as follows.

$$R = \begin{matrix} & \begin{matrix} 3 & 5 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \left[\begin{array}{ccc} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \end{matrix}$$

Operations on crisp relations

Suppose, $R(x, y)$ and $S(x, y)$ are the two relations defined over two crisp sets $x \in A$ and $y \in B$

- **Union:** $R(x, y) \cup S(x, y) = \max(R(x, y), S(x, y));$
- **Intersection:** $R(x, y) \cap S(x, y) = \min(R(x, y), S(x, y));$
- **Complement:** $\overline{R(x, y)} = 1 - R(x, y)$

Example: Operations on crisp relations

Suppose, $R(x, y)$ and $S(x, y)$ are the two relations defined over two crisp sets $x \in A$ and $y \in B$

$$R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } S = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the following

- $R \cup S$
- $R \cap S$
- $\bar{R}$

Composition of two crisp relations

Given R is a relation on X, Y and S is another relation on Y, Z . Then, $R \circ S$ is called a composition of relation on X and Z which is defined as follows.

$$R \circ S = \{(x, z) | (x, y) \in R \text{ and } (y, z) \in S \text{ and } \forall y \in Y\}$$

Max-Min Composition

Given the two relation matrices R and S , the **max-min composition** is defined as $T = R \circ S$;

$$T(x, z) = \max\{\min\{R(x, y), S(y, z) \text{ and } \forall y \in Y\}\}$$

Composition: Composition

Example : Given $X = \{1, 3, 5\}; Y = \{1, 3, 5\}; R = \{(x, y) | y = x + 2\};$
 $S = \{(x, y) | x < y\}$

Here, R and S is on $X \times Y$.

Thus, we have $R = \{(1, 3), (3, 5)\}, S = \{(1, 3), (1, 5), (3, 5)\}$

$$R = \begin{matrix} & \begin{matrix} 1 & 3 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad \text{and} \quad S = \begin{matrix} & \begin{matrix} 1 & 3 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Using max-min composition

$$R \circ S = \begin{matrix} & \begin{matrix} 1 & 3 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

1 3 5

Fuzzy relations

- Fuzzy relation is a fuzzy set defined on the Cartesian product of crisp set $X_1, X_2, \dots, X_n$
- Here, n-tuples $(x_1, x_2, \dots, x_n)$ may have varying degree of memberships within the relationship.
- The membership values indicate the strength of the relation between the tuples.

Fuzzy relations

Example:

$$X = \{ \text{typhoid}, \text{viral}, \text{cold} \}, Y = \{ \text{running nose}, \text{high temp}, \text{shivering} \}$$

The fuzzy relation R is defined as

$$R = \begin{matrix} & \begin{matrix} \text{typhoid} \\ \text{viral} \\ \text{cold} \end{matrix} & \begin{bmatrix} 0.1 & 0.9 & 0.8 \\ 0.2 & 0.9 & 0.7 \\ 0.9 & 0.4 & 0.6 \end{bmatrix} \\ & \begin{matrix} \text{running nose} & \text{high temperature} & \text{shivering} \end{matrix} \end{matrix}$$

Fuzzy Cartesian product

Suppose

- A is a fuzzy set on the universe of discourse X with $\mu_A(x) | x \in X$
- B is a fuzzy set on the universe of discourse Y with $\mu_B(y) | y \in Y$

Then $R = A \times B \subset X \times Y$; where R has its membership function given

by $\mu_R(x, y) = \mu_{A \times B}(x, y) = \min\{\mu_A(x), \mu_B(y)\}$

Fuzzy Cartesian product

Example :

$A = \{(a_1, 0.2), (a_2, 0.7), (a_3, 0.4)\}$ and $B = \{(b_1, 0.5), (b_2, 0.6)\}$

$$R = A \times B = \begin{matrix} & \begin{matrix} a_1 & a_2 & a_3 \end{matrix} \\ \begin{matrix} b_1 & b_2 \end{matrix} & \begin{bmatrix} 0.2 & 0.2 \\ 0.5 & 0.6 \\ 0.4 & 0.4 \end{bmatrix} \end{matrix}$$

Operations on Fuzzy relations

Let R and S be two fuzzy relations on $A \times B$.

- **Union:** $\mu_{R \cup S}(a, b) = \max\{\mu_R(a, b), \mu_S(a, b)\}$
- **Intersection:** $\mu_{R \cap S}(a, b) = \min\{\mu_R(a, b), \mu_S(a, b)\}$
- **Complement:** $\mu_{\bar{R}}(a, b) = 1 - \mu_R(a, b)$
- **Composition:** $T = R \circ S$
$$\mu_{R \circ S} = \max_{y \in Y} \{\min(\mu_R(x, y), \mu_S(y, z))\}$$

Operations on Fuzzy relations: Example

Example : $X = (x_1, x_2, x_3)$, $Y = (y_1, y_2)$, $Z = (z_1, z_2, z_3)$,

$$R = \begin{matrix} & \begin{matrix} x_1 & x_2 & x_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{matrix} \quad \text{and} \quad S = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.4 & 0.7 \\ 0.5 & 0.8 & 0.9 \end{bmatrix} \end{matrix}$$

$$R \circ S = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.5 & 0.4 & 0.5 \\ 0.5 & 0.8 & 0.9 \\ 0.6 & 0.6 & 0.7 \end{bmatrix} \end{matrix}$$

$$\mu_{R \circ S}(x_1, y_1) = \max\{\min(\mu_R(x_1, y_1), \mu_S(y_1, z_1)), \min(\mu_R(x_1, y_2), \mu_S(y_2, z_1))\}$$

$$= \max\{\min(0.5, 0.6), \min(0.1, 0.5)\} = \max\{0.5, 0.1\} = 0.5 \text{ and so on.}$$

Fuzzy relation : An example

Consider the following two sets P and D , which represent a set of paddy plants and a set of plant diseases. More precisely

$P = \{P_1, P_2, P_3, P_4\}$ a set of four varieties of paddy plants

$D = \{D_1, D_2, D_3, D_4\}$ of the four various diseases affecting the plants.

In addition to these, also consider another set $S = \{S_1, S_2, S_3, S_4\}$ be the common symptoms of the diseases.

Let, R be a relation on $P \times D$, representing which plant is susceptible to which diseases, which is stated as

$$R = \begin{matrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{matrix} \begin{bmatrix} 0.6 & 0.6 & 0.9 & 0.8 \\ 0.1 & 0.2 & 0.9 & 0.8 \\ 0.9 & 0.3 & 0.4 & 0.8 \\ 0.9 & 0.8 & 0.4 & 0.2 \end{bmatrix} \begin{matrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{matrix}$$

Fuzzy relation : An example

Also, consider T be the another relation on $D \times S$, which is given by

$$S = \begin{matrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{matrix} \begin{bmatrix} 0.1 & 0.2 & 0.7 & 0.9 \\ 1.0 & 1.0 & 1.4 & 0.6 \\ 0.0 & 0.0 & 0.5 & 0.9 \\ 0.9 & 1.0 & 0.8 & 0.2 \end{bmatrix} \begin{matrix} S_1 \\ S_2 \\ S_3 \\ S_4 \end{matrix}$$

Obtain the association of plants with the different symptoms of the disease using **max-min composition**.

Hint: Find $R \circ T$, and verify that

$$R \circ S = \begin{matrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{matrix} \begin{bmatrix} 0.8 & 0.8 & 0.8 & 0.9 \\ 0.8 & 0.8 & 0.8 & 0.9 \\ 0.8 & 0.8 & 0.8 & 0.9 \\ 0.8 & 0.8 & 0.7 & 0.9 \end{bmatrix} \begin{matrix} S_1 \\ S_2 \\ S_3 \\ S_4 \end{matrix}$$

Fuzzy relation : Another example

Let, $R = x$ is relevant to y

and $S = y$ is relevant to z

be two fuzzy relations defined on $X \times Y$ and $Y \times Z$, respectively, where $X = \{1,2,3\}$, $Y = \{\alpha, \beta, \gamma, \delta\}$ and $Z = \{a, b\}$. Assume that R and S can be expressed with the following relation matrices :

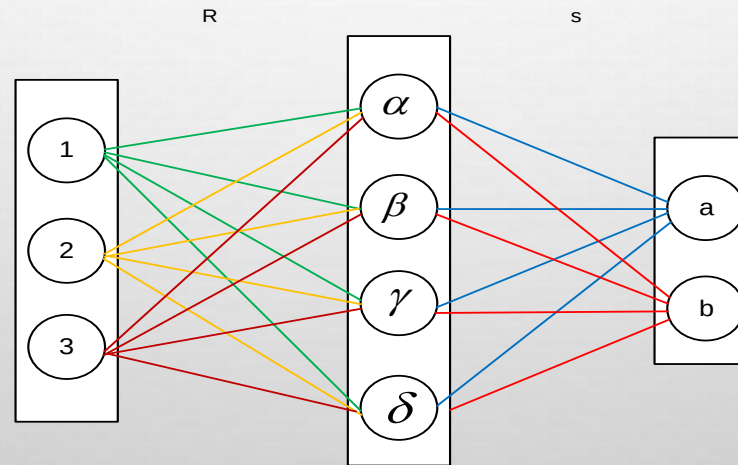
$$R = \begin{matrix} & \begin{matrix} \alpha & \beta & \gamma & \delta \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.1 & 0.3 & 0.5 & 0.7 \\ 0.4 & 0.2 & 0.8 & 0.9 \\ 0.6 & 0.8 & 0.3 & 0.2 \end{bmatrix} \end{matrix} \text{ and } S = \begin{matrix} & \begin{matrix} a & b \end{matrix} \\ \begin{matrix} \alpha \\ \beta \\ \gamma \\ \delta \end{matrix} & \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.3 \\ 0.5 & 0.6 \\ 0.7 & 0.2 \end{bmatrix} \end{matrix}$$

Fuzzy relation : Another example

Now, we want to find $R \circ S$, which can be interpreted as a derived fuzzy relation x is relevant to z .

Suppose, we are only interested in the degree of relevance between $2 \in X$ and $a \in Z$. Then, using max-min composition,

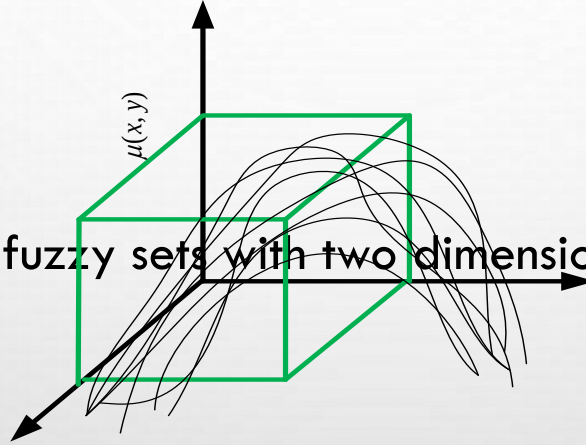
$$\begin{aligned}\mu_{R \circ S}(2, a) &= \max\{(0.4 \wedge 0.9), (0.2 \wedge 0.2), (0.8 \wedge 0.5), (0.9 \wedge 0.7)\} \\ &= \max\{0.4, 0.2, 0.5, 0.7\} = 0.7\end{aligned}$$



2D Membership functions : Binary fuzzy relations

(Binary) fuzzy relations are fuzzy sets $A \times B$ which map each element in $A \times B$ to a membership grade between 0 and 1 (both inclusive).

Note that a membership function of a binary fuzzy relation can be depicted with a 3D plot.



Important: Binary fuzzy relations are fuzzy sets with two dimensional MFs and so on.

2D membership function : An example

Let, $X = R^+ = y$ (the positive real line) and

$R = X \times Y = "y \text{ is much greater than } x"$

The membership function of $\mu_R(x, y)$ is defined as

$$\mu_R(x, y) = \begin{cases} \frac{(y - x)}{4} & \text{if } y > x \\ 0 & \text{if } y \leq x \end{cases}$$

Suppose, $X = \{3, 4, 5\}$ and $Y = \{3, 4, 5, 6, 7\}$, then

$$R = \begin{matrix} & \begin{matrix} 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0.25 & 0.5 & 0.75 & 1.0 \\ 0 & 0 & 0.25 & 0.5 & 0.75 \\ 0 & 0 & 0 & 0.25 & 0.5 \end{bmatrix} \end{matrix}$$

Example:

How you can derive the following?

If x is A or y is B then z is C ;

Given that

- $R1$: If x is A then z is C $[R1 \in A \times C]$
- $R2$: If y is B then z is C $[R2 \in B \times C]$

Hint:

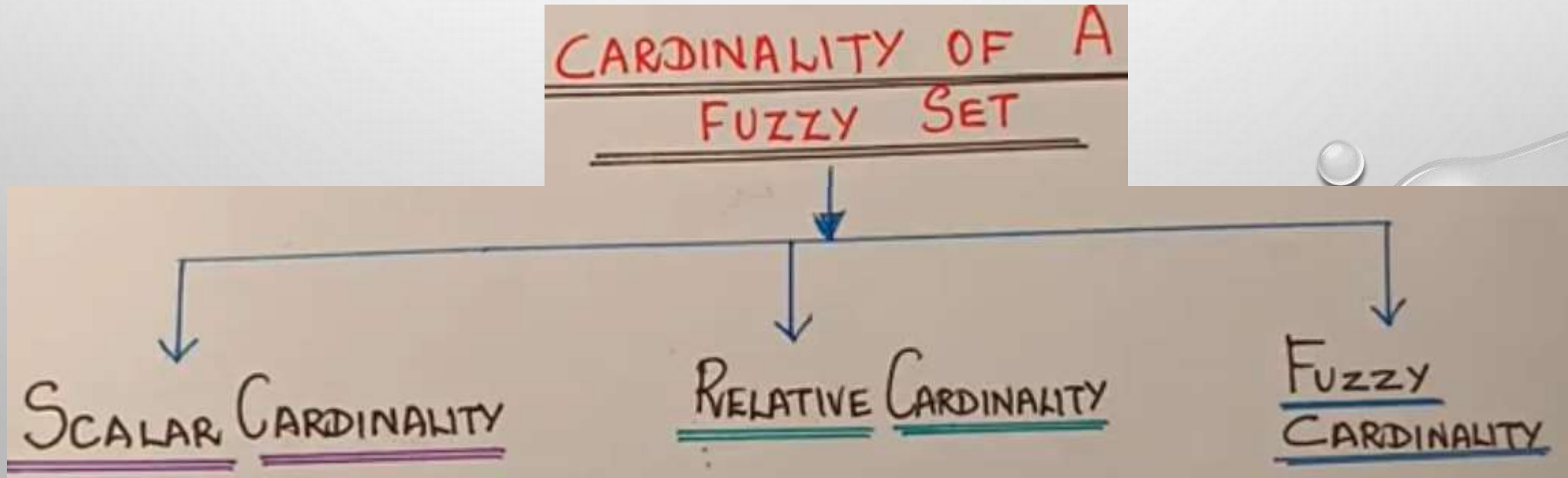
- ✓ You have given two relations $R1$ and $R2$.
- ✓ Then, the required can be derived using the **union operation** of $R1$ and $R2$

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Fuzzy cardinality

Fuzzy cardinality

Fuzzy cardinality extends the classical set cardinality to fuzzy sets by measuring the size of a set where elements have partial membership degrees, rather than just 0 or 1. It is primarily expressed as the **scalar cardinality** (sum of membership values, or sigma-count) or as a fuzzy set of potential natural numbers based on **alpha-cuts**



Fuzzy cardinality

Key Aspects of Fuzzy Cardinality:

Scalar Cardinality (Sigma-count): The most common approach, calculated by summing the membership grades $(\mu_A(x))$ of all elements in the fuzzy set A:

$$|A| = \sum_{x \in X} \mu_A(x)$$

Fuzzy Cardinality (Zadeh's Definition): Defines cardinality as a fuzzy set itself, representing a "fuzzy natural number". It is often computed by analyzing the scalar cardinality of various α -cuts (sets where $\mu_A(x) \geq \alpha$)

Relative Cardinality: Measures the ratio of the scalar cardinality to the total number of elements in the universe, often used for normalization